# The Grimoire

APPRENTICE ____________________

DATE ____________________

**What a trace does:** it runs a program with a pencil instead of a wand.

1. One row for every line of code, in the order Python runs them.
2. One gold column for every vessel (variable) in the program.
3. After each line, write what each vessel holds right now. Put a dash in a box that holds nothing yet.
4. When a line refills a vessel, **cross out the old value** and write the new one beside it. Never erase. The crossing-out is the point.

WORKED EXAMPLE · DONE FOR YOU

## The twice-filled vessel (Day 1)

| # | CODE | SPARKS vessel | OUTPUT |
|---|---|---|---|
| 1 | `sparks = 20` | 20 | — |
| 2 | `print(sparks)` | 20 | 20 |
| 3 | `sparks = 15` | ~~20~~ 15 | — |
| 4 | `print(sparks)` | 15 | 15 |

**Look hard at line 3.** The 20 gets crossed out. A vessel holds **one** value, and = pours: the old value is gone. After line 3, sparks is 15. Not 20, not both, not 35.

TRACE 1 · DAY 1, THE REGISTRATION SCROLL

## Sam registers with 3 tokens

The apprentice is **Sam** and they carry **3** spell tokens. Trace all four lines.

| # | CODE | NAME | TOKENS | SPARKS | OUTPUT |
|---|---|---|---|---|---|
| 1 | `name = input("...Apprentice: ")` | | | | |
| 2 | `tokens = int(input("...tokens? "))` | | | | |
| 3 | `sparks = tokens * 12` | | | | |
| 4 | `print(f"Scroll of Registration: ...")` | | | | |

TRACE 2 · DAY 1, THE LOOM

## One pass per row

Trace `for row in range(1, 5): print(row, "✦" * row)`. One table row per pass. **Put an X through any pass the loop never runs.**

| PASS | ROW<br>vessel | WHAT PRINTS, EXACTLY |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |

ORDERED PAIRS · DAYS 1–2, TWO RULES TOGETHER

## Corresponding patterns

| ROW | RUNES ("✦" * ROW) | RUNES ("✦" * (ROW + ROW)) |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |

The relationship between the two patterns, in my own words:

SHELF DIAGRAM · DAY 1, THE STOCKROOM

## Number every flask

Write one ingredient from your Gatekeeper shelf inside each flask, then write each flask's number on the line below it.

On this shelf, what happens when the program asks for `shelf[4]`, and why?

COORDINATES CORNER · DAY 3, WANDLIGHT

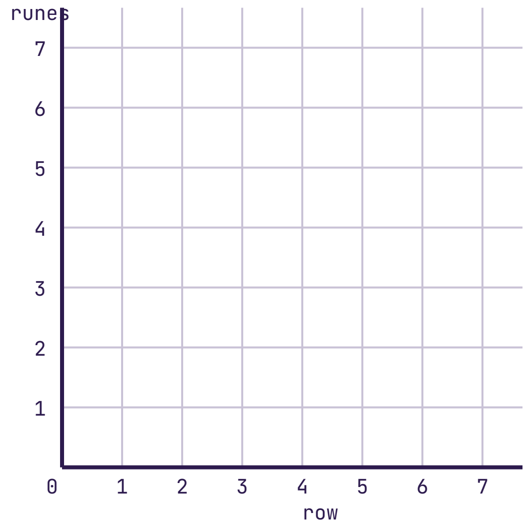
## Where the sparks live

The wand tip lives at ( ________ , ________ ).

As a spark rises, its y value gets ____________ because the screen counts y from the:

____________________________________________

Plot the staircase pairs (row, runes) on the grid: (1, 1) (2, 2) (3, 3) (4, 4). Stretch: plot the doubled pairs from your table too.




GRIMOIRE ENTRIES · ONE PER DAY, AT FIRST CASTING

## Exit tickets

| DAY | MY HINGE ANSWER | ONE MISFIRE I MET, AND THE ONE CHANGE THAT LIFTED IT |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |

Collected as evidence for **E5-PRO-VD-??** ⚠ unverified; confirm the code at source before distributing (Days 1–3 traces and shelf diagram), **5.OA.B.3** (ordered-pairs table and plotted grid), and **5.G.A.2** (coordinates corner), both mathematics codes verified via the Learning Commons CASE network.